Chapter 3 Multidimensional Random Variables

Section 2 Marginal and Conditional Distributions

Guo Yongjiang

Outline

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   - Examples (Continuous Marginals)
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Definition: Marginal Distribution Function

Definition: Marginal Distribution

Let $F(x, y)$ be the joint distribution function of (X, Y).
The function

$$F_X(x) = P(X \leq x) = F(x, +\infty) = \lim_{y \to +\infty} F(x, y)$$

is called the **marginal distribution function** of X.
Similarly,

$$F_Y(y) = P(Y \leq y) = F(+\infty, y) = \lim_{x \to +\infty} F(x, y)$$

is the **marginal distribution function** of Y.

Important Note: The joint distribution completely determines the marginals. However, the marginal distributions generally **cannot** uniquely determine the joint distribution. The joint distribution contains more information than the two marginals combined.

Discrete Case: Marginal PMF

If the joint PMF of (X, Y) is $p_{ij} = P(X = x_i,\ Y = y_j)$, then the marginal PMFs of X and Y are:

$$p_{i\cdot} = P(X = x_i) = \sum_{j=1}^{\infty} p_{ij}, \qquad p_{\cdot j} = P(Y = y_j) = \sum_{i=1}^{\infty} p_{ij}.$$

In the joint PMF table, marginal PMFs are simply the **row sums** and **column sums**.

$X \backslash Y$	y_1	y_2	$\cdots$	y_j	$p_{i\cdot}$
x_1	p_{11}	p_{12}	$\cdots$	p_{1j}	$p_{1\cdot}$
x_2	p_{21}	p_{22}	$\cdots$	p_{2j}	$p_{2\cdot}$
$\vdots$	$\vdots$	$\vdots$		$\vdots$	$\vdots$
x_i	p_{i1}	p_{i2}	$\cdots$	p_{ij}	$p_{i\cdot}$
$\vdots$	$\vdots$	$\vdots$		$\vdots$	$\vdots$
$p_{\cdot j}$	$p_{\cdot 1}$	$p_{\cdot 2}$	$\cdots$	$p_{\cdot j}$	1

Example: Marginal PMF (Question)

Example: Marginal Distribution of Discrete RV

The joint PMF of (X, Y) is given below. Find the marginal PMFs of X and Y.

$X \backslash Y$	0	1	2	$p_{i\cdot}$
0	0.10	0.15	0.05	0.30
1	0.20	0.25	0.10	0.55
2	0.05	0.05	0.05	0.15
$p_{\cdot j}$	0.35	0.45	0.20	1.00

Analysis: The marginal PMF is obtained by summing the rows (for X) and columns (for Y).

Example: Marginal PMF (Solution)

Solution

Row Sums for X:

$$P(X=0) = 0.10 + 0.15 + 0.05 = 0.30,$$
$$P(X=1) = 0.20 + 0.25 + 0.10 = 0.55,$$
$$P(X=2) = 0.05 + 0.05 + 0.05 = 0.15.$$

Column Sums for Y:

$$P(Y=0) = 0.10 + 0.20 + 0.05 = 0.35,$$
$$P(Y=1) = 0.15 + 0.25 + 0.05 = 0.45,$$
$$P(Y=2) = 0.05 + 0.10 + 0.05 = 0.20.$$

(Check: $\sum p_{i\cdot} = 1$ and $\sum p_{\cdot j} = 1$, satisfying normalization.)

Proposition: Marginal Density Formula

Proposition

If the joint density of (X, Y) is $f(x, y)$, then the marginal densities of X and Y are

$$f_X(x) = \int_{-\infty}^{+\infty} f(x, y)\, dy, \qquad f_Y(y) = \int_{-\infty}^{+\infty} f(x, y)\, dx.$$

Proof:

$F_X(x) = P(X \leq x) = P(X \leq x, Y < +\infty) = \lim_{y \to +\infty} F(x, y)$.

That is, $F_X(x) = \int_{-\infty}^{x} \int_{-\infty}^{+\infty} f(u, v)\, dv\, du$.

Taking the derivative with respect to x yields $f_X(x)$. The proof for Y follows similarly.

Example: Uniform on Unit Square (Question)

Example: Marginal Densities on Unit Square

Let the joint density of (X, Y) be

$$f(x, y) = \begin{cases} 1, & 0 \le x \le 1,\ 0 \le y \le 1, \\ 0, & \text{otherwise.} \end{cases}$$

Find the marginal densities $f_X(x)$ and $f_Y(y)$.

Analysis:
For $0 \le x \le 1$, integrate y from 0 to 1 (since the support is a unit square). Outside this range, the marginal density is 0.

Example: Uniform on Unit Square (Solution)

Solution

For $0 \leq x \leq 1$:

$$f_X(x) = \int_0^1 1 \, dy = 1.$$

For $x < 0$ or $x > 1$: $f_X(x) = 0$.
Thus,

$$f_X(x) = \begin{cases} 1, & 0 \leq x \leq 1, \\ 0, & \text{otherwise.} \end{cases}$$

Similarly, $f_Y(y) = 1$ for $0 \leq y \leq 1$, and 0 otherwise.
We see $X \sim U(0,1)$ and $Y \sim U(0,1)$.

Example: Non-Rectangular Support (Question)

Example: Marginal Densities with Curved Support

Let the joint density of (X, Y) be

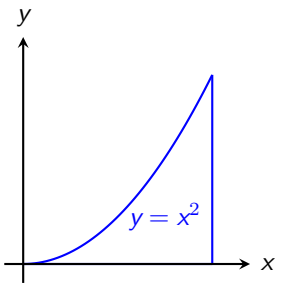
$$f(x, y) = \begin{cases} c\,x, & 0 \le x \le 1,\ 0 \le y \le x^2, \\ 0, & \text{otherwise}, \end{cases}$$

where $c > 0$ is a constant. Find c, $f_X(x)$, and $f_Y(y)$.

Example: Non-Rectangular Support (Analysis)

Analysis:

- **Find** c**:** use normalization $\iint f(x,y)\,dx\,dy = 1$.
- **Find** $f_X(x)$**:** integrate y from 0 to x^2.
- **Find** $f_Y(y)$**:** the range is $y \le x^2 \Rightarrow x \ge \sqrt{y}$, so integrate x from $\sqrt{y}$ to 1.



Example: Non-Rectangular Support (Solution)

Solution

Find c:

$$\int_0^1 \int_0^{x^2} cx\,dy\,dx = c\int_0^1 x(x^2)\,dx = \frac{c}{4} = 1 \implies c = 4.$$

Marginal $f_X(x)$: For $0 \le x \le 1$,

$$f_X(x) = \int_0^{x^2} 4x\,dy = 4x^3.$$

Marginal $f_Y(y)$: For $0 \le y \le 1$,

$$f_Y(y) = \int_{\sqrt{y}}^1 4x\,dx = 2x^2\Big|_{\sqrt{y}}^1 = 2(1-y).$$

Note: The limits for $f_Y(y)$ depend on the non-rectangular support.

Definition: Conditional PMF

Conditional PMF

For discrete (X, Y), if $P(Y = y_j) > 0$, then

$$P(X = x_i \mid Y = y_j) = \frac{P(X = x_i,\ Y = y_j)}{P(Y = y_j)} = \frac{p_{ij}}{p_{\cdot j}}$$

is the **conditional PMF** of X given $Y = y_j$.
Similarly, if $P(X = x_i) > 0$,

$$P(Y = y_j \mid X = x_i) = \frac{p_{ij}}{p_{i\cdot}}.$$

Example: Conditional PMF (Question)

Example: Conditional Distribution of Discrete RV

Using the joint PMF from the previous example, find the conditional distribution of X given $Y = 1$, and Y given $X = 1$.

Analysis:

We already have $P(Y = 1) = 0.45$ and $P(X = 1) = 0.55$.

Divide the joint probabilities in the column $Y = 1$ by 0.45, and the row $X = 1$ by 0.55.

Example: Conditional PMF (Solution)

Solution

Given $Y = 1$:

$$P(X = 0 \mid Y = 1) = \frac{0.15}{0.45} = \frac{1}{3}, \quad P(X = 1 \mid Y = 1) = \frac{0.25}{0.45} = \frac{5}{9},$$
$$P(X = 2 \mid Y = 1) = \frac{0.05}{0.45} = \frac{1}{9}.$$

Given $X = 1$:

$$P(Y = 0 \mid X = 1) = \frac{0.20}{0.55} = \frac{4}{11}, \quad P(Y = 1 \mid X = 1) = \frac{0.25}{0.55} = \frac{5}{11},$$
$$P(Y = 2 \mid X = 1) = \frac{0.10}{0.55} = \frac{2}{11}.$$

(Both sum to 1.)

Definition: Conditional Density

Conditional Density

For continuous (X, Y), if $f_Y(y) > 0$, then

$$f_{X|Y}(x \mid y) = \frac{f(x, y)}{f_Y(y)}$$

is the **conditional density** of X given $Y = y$.
Similarly, if $f_X(x) > 0$,

$$f_{Y|X}(y \mid x) = \frac{f(x, y)}{f_X(x)}.$$

Derivation: From Conditional CDF to Density

Since $P(Y = y) = 0$ for continuous Y, define via limit as $h \to 0^+$:

$$P(X \le x \mid Y = y) = \lim_{h \to 0^+} P(X \le x \mid y < Y \le y + h).$$

Using conditional probability and joint density:

$$= \lim_{h \to 0^+} \frac{P(X \le x,\ y < Y \le y + h)}{P(y < Y \le y + h)} = \lim_{h \to 0^+} \frac{\int_{-\infty}^{x} \int_{y}^{y+h} f(u, v)\, dv\, du}{\int_{y}^{y+h} f_Y(v)\, dv}.$$

By the Mean Value Theorem (as $h \to 0$):

$$= \frac{\int_{-\infty}^{x} f(u, y)\, du}{f_Y(y)} = \int_{-\infty}^{x} \frac{f(u, y)}{f_Y(y)}\, du.$$

The integrand is exactly the conditional density $f_{X|Y}(x \mid y)$.

Example: Unit Square Conditional Density (Question and Analysis)

Example: Conditional Densities on Unit Square

Let (X, Y) have the joint density

$$f(x,y) = \begin{cases} 1, & 0 \le x \le 1,\ 0 \le y \le 1, \\ 0, & \text{otherwise.} \end{cases}$$

Find the conditional densities $f_{X|Y}(x \mid y)$ and $f_{Y|X}(y \mid x)$.

Analysis:

From previous examples, the marginal densities are:

$$f_X(x) = \begin{cases} 1, & 0 \le x \le 1, \\ 0, & \text{otherwise,} \end{cases} \qquad f_Y(y) = \begin{cases} 1, & 0 \le y \le 1, \\ 0, & \text{otherwise.} \end{cases}$$

Example: Unit Square Conditional Density (Formulas to Use)

Conditional Density Formulas

We use the definitions of conditional densities to compute the answers:

- **Given $Y = y$, the conditional density of X is:**

$$f_{X|Y}(x \mid y) = \frac{f(x,y)}{f_Y(y)}.$$

- **Given $X = x$, the conditional density of Y is:**

$$f_{Y|X}(y \mid x) = \frac{f(x,y)}{f_X(x)}.$$

Next Step: Substitute the joint density $f(x,y)$ and the marginal densities $f_X(x), f_Y(y)$ into these formulas.

Example: Unit Square Conditional Density (Solution)

Solution for $f_{X|Y}(x \mid y)$

Given $0 \le y \le 1$, the marginal $f_Y(y) = 1$. For $0 \le x \le 1$:

$$f_{X|Y}(x \mid y) = \frac{f(x,y)}{f_Y(y)} = \frac{1}{1} = 1.$$

Complete piecewise form:

$$f_{X|Y}(x \mid y) = \begin{cases} 1, & 0 \le x \le 1, \\ 0, & \text{otherwise.} \end{cases}$$

Thus, given $Y = y$, $X \sim U(0,1)$.

Solution for $f_{Y|X}(y \mid x)$

Example: Curved Support Conditional Density (Question)

Example: Conditional Density with Non-Rectangular Support

Let the joint density of (X, Y) be

$$f(x, y) = \begin{cases} 4x, & 0 \le x \le 1,\ 0 \le y \le x^2, \\ 0, & \text{otherwise.} \end{cases}$$

Find the conditional density $f_{X|Y}(x \mid y)$ for $0 \le y \le 1$.

Analysis:
From the previous calculations, the marginal density of Y is $f_Y(y) = 2(1 - y)$ for $0 \le y \le 1$. On the support region $0 \le y \le x^2 \le 1$, we have $x \ge \sqrt{y}$, which means x is restricted to the interval $[\sqrt{y}, 1]$.

Example: Curved Support Conditional Density (Solution)

Solution

For $0 \le y \le 1$ and $\sqrt{y} \le x \le 1$:

$$f_{X|Y}(x \mid y) = \frac{4x}{2(1-y)} = \frac{2x}{1-y}.$$

Outside this interval, it is 0.
(Check: $\int_{\sqrt{y}}^{1} \frac{2x}{1-y}\,dx = \frac{1}{1-y}x^2\Big|_{\sqrt{y}}^{1} = 1.$)

Remark: The support of the conditional density $[\sqrt{y}, 1]$ depends on y, indicating that X and Y are not independent.

Example: Reversing Conditional Densities (Question)

Example: Given Marginal and Conditional Densities

Let the marginal density of Y be

$$f_Y(y) = \begin{cases} 3y^2, & 0 \le y \le 1, \\ 0, & \text{otherwise.} \end{cases}$$

Given $Y = y$ (for $0 < y \le 1$), the conditional density of X is

$$f_{X|Y}(x \mid y) = \begin{cases} \dfrac{2x}{y^2}, & 0 \le x \le y, \\ 0, & \text{otherwise.} \end{cases}$$

Find the reverse conditional density $f_{Y|X}(y \mid x)$ for $0 < x \le 1$.

Example: Reversing Conditional Densities (Solution Part 1)

Step 1: Construct the Joint Density

By the multiplication rule for conditional densities:

$$f(x, y) = f_Y(y) \cdot f_{X|Y}(x \mid y).$$

Substituting the given densities, for $0 \leq y \leq 1$ and $0 \leq x \leq y$:

$$f(x, y) = (3y^2) \cdot \left(\frac{2x}{y^2}\right) = 6x.$$

Outside the region $0 \leq x \leq y \leq 1$, $f(x, y) = 0$.

Example: Reversing Conditional Densities (Solution Part 2)

Step 2: Find the Marginal Density of X

The support region is $0 \leq x \leq y \leq 1$. For a fixed $x \in (0, 1]$, the variable y ranges from x to 1. Integrating out y:

$$f_X(x) = \int_{-\infty}^{+\infty} f(x, y)\, dy = \int_x^1 6x\, dy = 6x(1 - x), \quad 0 \leq x \leq 1.$$

Outside this range, $f_X(x) = 0$.

Example: Reversing Conditional Densities (Solution Part 3)

Step 3: Find the Reverse Conditional Density

Using the definition of conditional density, we divide the joint density by the marginal density of X:

$$f_{Y|X}(y \mid x) = \frac{f(x,y)}{f_X(x)} = \frac{6x}{6x(1-x)} = \frac{1}{1-x}.$$

Recall the support region: given $X = x$, y must satisfy $x \leq y \leq 1$. Therefore, the complete conditional density is:

$$f_{Y|X}(y \mid x) = \begin{cases} \dfrac{1}{1-x}, & x \leq y \leq 1, \\ 0, & \text{otherwise.} \end{cases}$$

Example: Reversing Conditional Densities (Remark)

Key Takeaways

1. **Joint Density Construction:** Use the multiplication rule $f(x, y) = f_Y(y) f_{X|Y}(x \mid y)$ to combine the known marginals and conditionals.
2. **Integration Limits:** The support $0 \leq x \leq y \leq 1$ implies $y \in [x, 1]$ when integrating to find $f_X(x)$. The dynamic limits are crucial here.
3. **Independence Check:** The resulting conditional density $f_{Y|X}(y \mid x) = \frac{1}{1-x}$ depends on x, and its support $[x, 1]$ depends on x. This confirms that X and Y are **not independent**.

Example: Bivariate Normal Conditional Distribution

Theorem

Let $(X, Y) \sim N(\mu_1, \mu_2, \sigma_1^2, \sigma_2^2, \rho)$. Then:

$$X \mid Y = y \;\sim\; N\left(\mu_1 + \rho\frac{\sigma_1}{\sigma_2}(y - \mu_2),\; \sigma_1^2(1 - \rho^2)\right),$$

$$Y \mid X = x \;\sim\; N\left(\mu_2 + \rho\frac{\sigma_2}{\sigma_1}(x - \mu_1),\; \sigma_2^2(1 - \rho^2)\right).$$

Proof: Completing the Square

Focus on the exponent of the joint normal density. Let $u = \frac{x-\mu_1}{\sigma_1}$, $v = \frac{y-\mu_2}{\sigma_2}$.
The exponential part is $Q(x,y) = u^2 - 2\rho uv + v^2$.
Complete the square in v:

$$Q(x,y) = (v - \rho u)^2 + (1-\rho^2)u^2.$$

Substitute back:

$$\frac{Q}{2(1-\rho^2)} = \frac{1}{2(1-\rho^2)}\left(\frac{y-\mu_2}{\sigma_2} - \rho\frac{x-\mu_1}{\sigma_1}\right)^2 + \frac{(x-\mu_1)^2}{2\sigma_1^2}.$$

The joint density factors into:

$$f(x,y) = \underbrace{\frac{1}{\sqrt{2\pi}\sigma_1} e^{-\frac{(x-\mu_1)^2}{2\sigma_1^2}}}_{f_X(x)} \cdot \underbrace{\frac{1}{\sqrt{2\pi}\sigma_2\sqrt{1-\rho^2}} e^{-\frac{1}{2(1-\rho^2)}\left(\frac{y-\mu_2}{\sigma_2} - \rho\frac{x-\mu_1}{\sigma_1}\right)^2}}_{f_{Y|X}(y|x)}.$$

The second factor integrates to 1 over y, so it is exactly the conditional density.

Proof: Identifying the Normal Parameters

The conditional density is:

$$f_{Y|X}(y \mid x) = \frac{1}{\sqrt{2\pi}\sigma_2\sqrt{1-\rho^2}} \exp\left\{ -\frac{\left(y - \mu_2 - \rho\frac{\sigma_2}{\sigma_1}(x-\mu_1)\right)^2}{2\sigma_2^2(1-\rho^2)} \right\}.$$

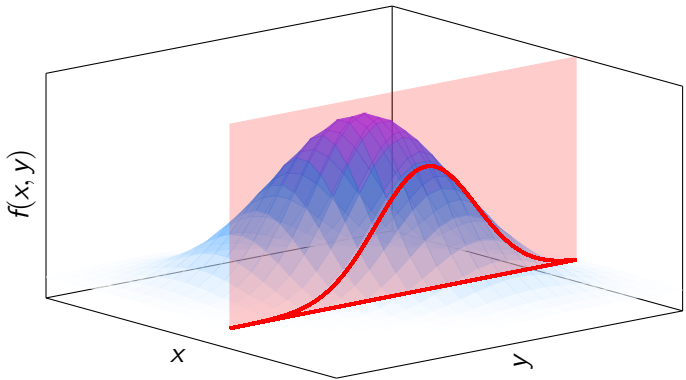
Compare with the standard normal density $\frac{1}{\sqrt{2\pi}\sigma} \exp\left\{-\frac{(y-\mu)^2}{2\sigma^2}\right\}$:

$$\mu = \mu_2 + \rho\frac{\sigma_2}{\sigma_1}(x-\mu_1), \qquad \sigma^2 = \sigma_2^2(1-\rho^2).$$

Therefore,

$$Y \mid X = x \sim N\left(\mu_2 + \rho\frac{\sigma_2}{\sigma_1}(x-\mu_1),\ \sigma_2^2(1-\rho^2)\right).$$

Visualizing the Conditional Distribution



The red curve shows the conditional density $Y \mid X = x_0$, which remains a normal bell curve.

Section Summary

- **Marginal Distribution**: Obtained by summing (discrete) or integrating (continuous) over the other variable. Joint distribution determines marginals, but not vice versa.
- **Conditional Distribution**: The distribution of one variable given the other is fixed.
- **Conditional Density Formula**: $f_{X|Y}(x \mid y) = \frac{f(x,y)}{f_Y(y)}$. Be careful with the integration limits, especially for non-rectangular support.
- **Bivariate Normal**: Conditional distributions remain normal, with a mean that shifts linearly (foundation of linear regression).